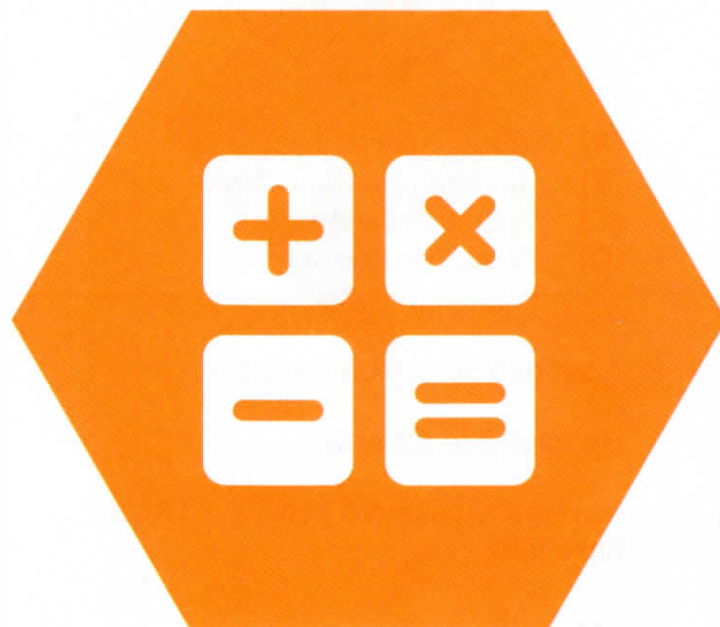




UNSW Global
AUSTRALIA

PAPER B



2015 ICAS

International Competitions
and Assessments for Schools

MATHEMATICS

Educational Assessment Australia
eaa.unsw.edu.au

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.

You are **NOT** allowed to use a calculator.

1. What time is shown on this clock?



- (A) three minutes past eight
(B) twenty minutes to three
(C) a quarter past eight
(D) a quarter to eight

2. Allan had 30 coloured pencils.

He gave 8 to his sister.

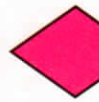
How many coloured pencils did Allan have left?

- (A) 8
(B) 12
(C) 22
(D) 38

3. Adam is making a pattern using shapes.



What must the next shape be in Adam's pattern?

- (A)  (B)  (C)  (D) 

4. Lori has a spot of paint on the right side of her face below her eye.

Which of these is a picture of Lori?



(A)



(B)



(C)



(D)

5. Fadi is planning a party for 16 people. He wants to buy party hats for them.

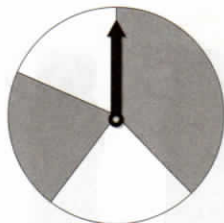


Party hats are sold in packets of five hats. Fadi wants everyone to have a party hat.

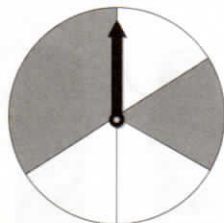
What is the smallest number of packets he must buy?

- (A) 3
(B) 4
(C) 11
(D) 16

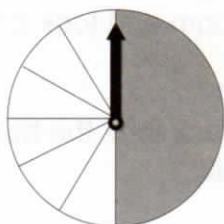
6. Which of the following spinners has more chance of landing on white than on grey?



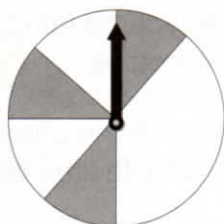
(A)



(B)



(C)

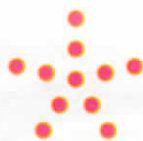


(D)

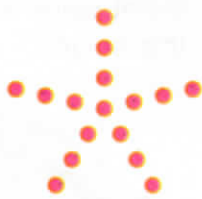
7. Vijay drew a pattern of shapes made up of dots.



Shape 1



Shape 2



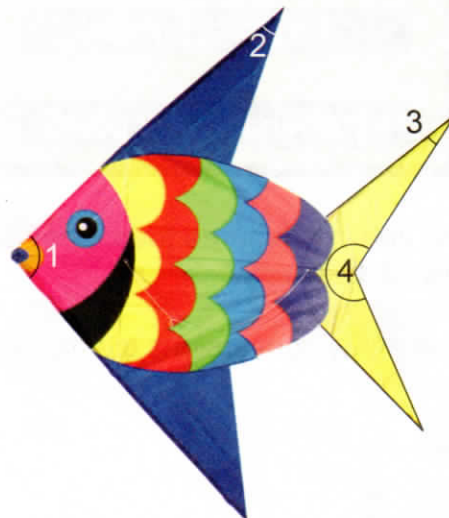
Shape 3

How many dots will be in Shape 4?

- (A) 21
(B) 24
(C) 25
(D) 27

8. Leila had a kite.

She measured some of the angles on the kite.



What is the order of the angles from smallest to largest?

- (A) 3, 2, 1, 4
(B) 3, 2, 4, 1
(C) 2, 3, 1, 4
(D) 2, 3, 4, 1

9. Manasa made this solid.



He then painted three of the faces red and the rest blue.

How many faces did he paint blue?

- (A) 1
(B) 4
(C) 5
(D) 8

10. The Jets played the Knights in a game of rugby in which only tries and conversions were scored. This is the scoring system.

1 try = 5 points

1 conversion = 2 points

The Jets made three tries. They scored a total of 19 points.

How many conversions did the Jets make?

- (A) 1
- (B) 2
- (C) 4
- (D) 7

11. Jane is counting up by threes.

She starts to write this pattern in four columns as shown.

(A)	(B)	(C)	(D)
3	6	9	12
15	18	21	24
27			

In which column should Jane write 42?

12. Sally filled four black bottles with cordial. Three bottles had orange cordial and one had lime cordial.



Jack came into the room and took one of the bottles.

What is the chance Jack took the bottle filled with lime cordial?

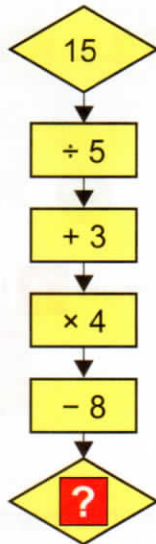
- (A) unlikely
- (B) likely
- (C) certain
- (D) even chance

13. What value is the arrow pointing to on this scale?



- (A) 60
- (B) 59
- (C) 54
- (D) 47

14. Billy started with the number 15 and followed the instructions.



What number did he finish with?

15. Polly's scoop holds 80 grams of flour. She fills her jar with 5 full scoops of flour.



How much more flour does the jar hold than the scoop?

16. Fadi drew this shape.



Which of these shows Fadi's shape flipped and turned?



(A)



(B)

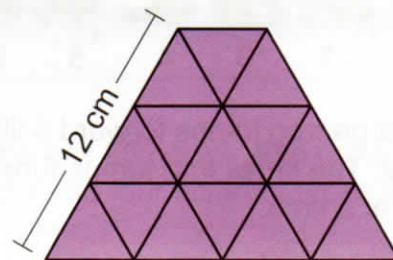


(C)



(D)

17. What is the perimeter of this shape, in centimetres?



- (A) 11
(B) 15
(C) 33
(D) 44

18. Darsh delivers furniture with his father.

This table shows how much money they earned over five days.

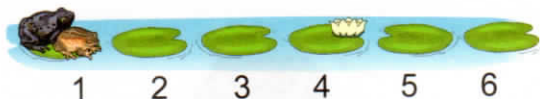
Monday	   
Tuesday	 
Wednesday	 
Thursday	    
Friday	    

Key:  = \$100

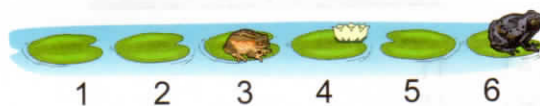
How much more money did they earn on Thursday than on Tuesday?

- (A) \$650
(B) \$450
(C) \$350
(D) \$150

19. Two frogs sit on a lily pad as shown.



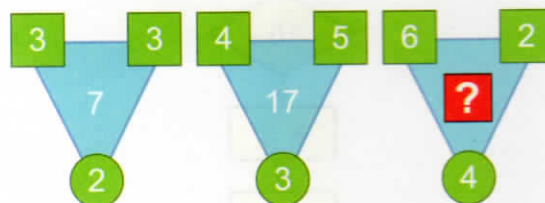
The large frog jumps forward 5 lily pads at a time. The small frog jumps forward 2 lily pads at a time.



What is the number of the first lily pad that both frogs will land on?

- (A) 7
(B) 9
(C) 10
(D) 11

20. The numbers in each diagram follow the same pattern.



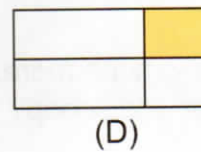
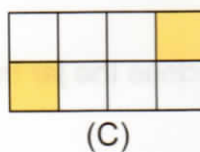
What number must  be?

- (A) 4
(B) 8
(C) 26
(D) 27

21. $15 + \text{?} = 32 - 6$

- (A) 11
(B) 17
(C) 23
(D) 26

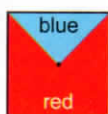
22. Which shape does **NOT** show one-quarter shaded?



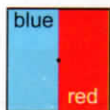
23. Four children each made a square out of cardboard. All the squares had the same area. Each child used two pieces of cardboard, one red and one blue.



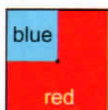
Anna



Mia



Raj



Danny

Which statement is true?

- (A) Raj's red piece has the same area as Danny's red piece.
- (B) Mia's blue piece has the same area as Anna's blue piece.
- (C) Anna's red piece has the same area as Danny's red piece.
- (D) Mia's blue piece has the same area as Danny's blue piece.

24. Greg weighed 9 identical buttons.
They weighed 27 grams.



How much will 5 buttons weigh?

- (A) 9 grams
- (B) 13 grams
- (C) 15 grams
- (D) 20 grams

25. Faye thinks of a four-digit number.

- The thousands digit is 3 times the tens digit.
- The hundreds digit is even.
- The units digit is an odd number less than 5.

Which of the following could be Faye's number?

- (A) 1933
- (B) 2663
- (C) 6425
- (D) 9831

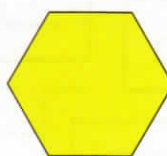
26. Which of these shapes is a pentagon?



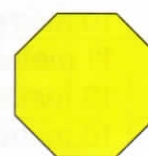
(A)



(B)



(C)



(D)

27. Fiona has 25 books. She grouped the books into four different types as shown in the table.

	Fiction	Nonfiction
Pictures	7	8
No Pictures	?	4

How many fiction books have no pictures?

- (A) 3
- (B) 6
- (C) 13
- (D) 19

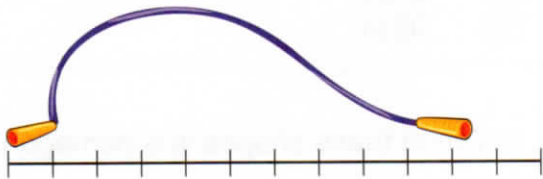
28. Yara and Danny collected a total of 48 shells.

Danny collected 3 times as many shells as Yara did.

How many shells did Yara collect?

- (A) 12
- (B) 16
- (C) 32
- (D) 36

29.



$\text{—|—} = 1 \text{ metre}$

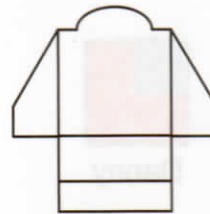
Which of these is closest to the length of the skipping rope, including the handles?

- (A) 10 metres
- (B) 11 metres
- (C) 13 metres
- (D) 16 metres

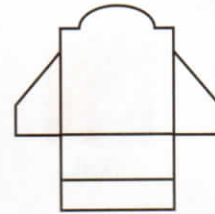
30. This is a display box.



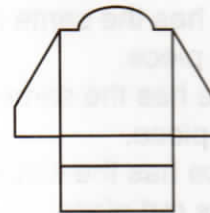
Which of the following is a net of this box?



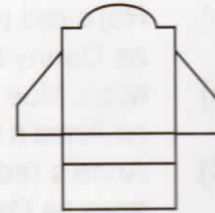
(A)



(B)



(C)



(D)

31. Dedi had a combination lock. To open the lock he started at 37.

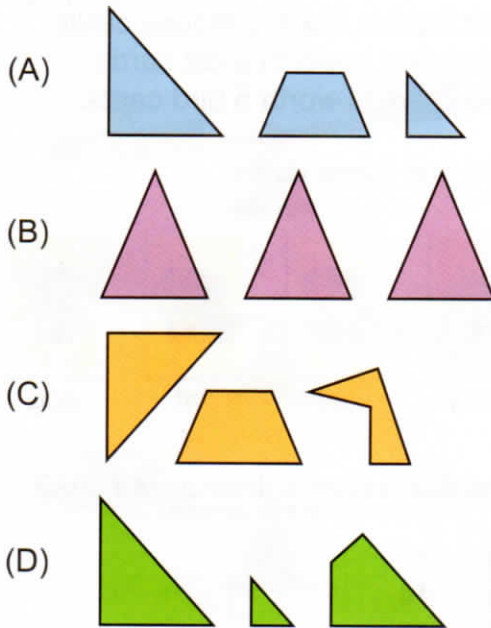
He made three half turns clockwise and three quarter turns anticlockwise.



What number did he end up at?

- (A) 7
- (B) 17
- (C) 27
- (D) 37

32. Which option shows three shapes that can be arranged to make one triangle?



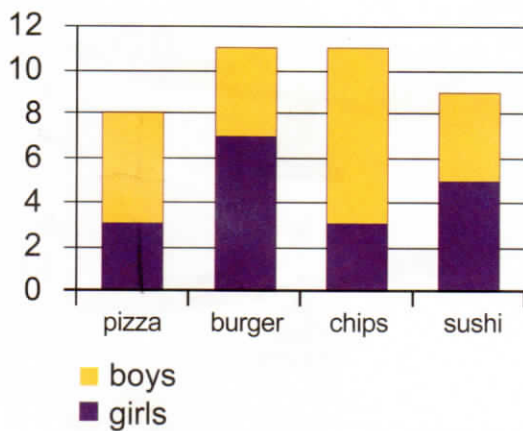
34. Karen read this sign as she reached the train station.



For how many hours is track work being carried out this weekend?

- (A) 16
(B) 28
(C) 34
(D) 40

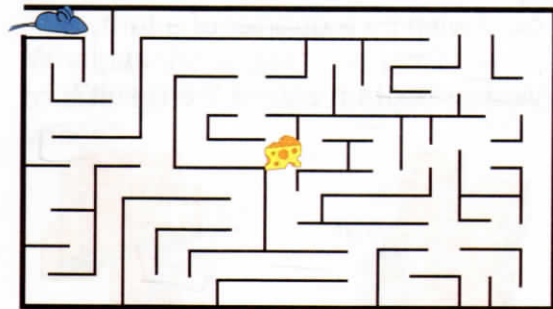
33. Louise asked her classmates to name their favourite takeaway food. The graph below shows this information.



How many of Louise's classmates are girls?

- (A) 9
(B) 18
(C) 21
(D) 39

35. A mouse enters a maze looking for cheese.



What is the least number of quarter turns the mouse can make to reach the cheese?

- (A) 6
(B) 7
(C) 11
(D) 15

36. Danny uses a cup and a jug to mix juice. It takes 2 cups of juice to fill the jug. He pours orange juice into the jug until it is half full. Then he adds half a cup of apple juice.

How full is the jug now?

- (A) one-quarter full
- (B) two-quarters full
- (C) three-quarters full
- (D) four-quarters full

37. Sharon made this solid from seven cubes.



Keeping the cube marked with the white symbol facing her, she rotated the solid clockwise by a quarter of a turn.

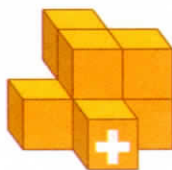
Which diagram shows the result?



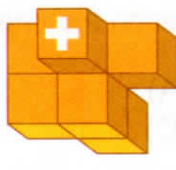
(A)



(B)



(C)



(D)

38. Jess collects animal cards.

1 mouse card is worth 1 point.
1 cat card is worth 3 mouse cards.
1 bird card is worth 2 cat cards.
1 dog card is worth 5 bird cards.

Jess has these cards.



mouse



cat



cat

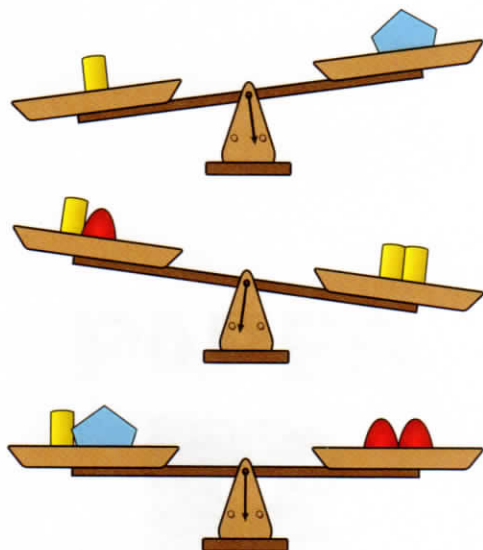


dog

How many points does Jess have?

- (A) 12
- (B) 14
- (C) 17
- (D) 37

39. Ravi used a balance to compare the masses of some objects. His results are shown below.



Which of the following must be true?

- (A)
- (B)
- (C)
- (D)

40. Sue has four flowers. She can arrange her flowers equally between vases in three different ways.

(1) one vase – four flowers



(2) two vases – two flowers in each



(3) four vases – one flower in each



Raj, Danny, Haji and Edward have some flowers.

Who has the largest number of ways to arrange his flowers equally between vases?

Edward
17 flowers



(A)

Haji
16 flowers



(B)

Danny
15 flowers



(C)

Raj
14 flowers



(D)

2015	Paper B		Maths answer		
1	C		21	A	
2	C		22	D	
3	A		23	D	
4	A		24	C	
5	B		25	D	
6	D		26	A	
7	A		27	B	
8	A		28	A	
9	C		29	C	
10	B		30	B	
11	B		31	A	
12	A		32	D	
13	C		33	B	
14	C		34	D	
15	C		35	B	
16	A		36	C	
17	D		37	B	
18	C		38	D	
19	D		39	A	
20	B		40	B	